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Keywords
Agriculture, Inequality, Climate Change, Conflict

JEL Codes
D74, Q54, Q18, F51

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Crop and Conflict: exploring the impact of Inequality in Agricultural Production on Conflict Risk

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Abstract

Recent attempts to find a robust empirical correlation between climate variability, crop production and civil conflict risk have been quite inconclusive. In this paper, we argue that the ambiguity in empirical findings may be partly due to a general tendency to treat agricultural production in absolute terms, while neglecting the importance of the relative deprivation triggered by unequal distribution in crop yields across locations and between groups. To test this hypothesis, we rely on high-resolution global gridded data on the local yield of four main crops for the period 1981-2017, and calculate the level of inequality in crop production by means of a Gini index using the grid-cell information on yearly crop yields both at the country level and between identity based groups (regional and ethnic). In addition, for each level of spatial disaggregation, we compute the Gini coefficient using information on crop production from rural grid-cells only. Our results reveal a strong and robust association between crop inequality and the probability of conflict outbreak. This effect is particularly pronounced in the case of inequality among rural areas. Climatic variability, on the other hand, is shown not to increase significantly the destabilizing effect of crop inequality. We also find that akin ethnic competition and discrimination along ethnic lines represent good predictors of conflict outbreak, especially in the case of ethnic conflicts, and when the between-group inequality in crop production is calculated along ethnic boundaries.

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1 Introduction

Recent years have witnessed an upsurge in both the number and severity of conflicts (Dupuy and Rustad, 2018). Although the trend seems to be reversing at the present time, the rise in non-state conflicts and new emerging threats akin the growth of the Islamic State (Pettersson and Eck, 2017), suggest that the shadow of violence is not likely to vanish (Hegre et al., 2016) and will continue to lead to massive displacement of people, increased poverty, and higher undernourishment (Gates et al., 2012).

Contributing to the long debate on the root causes of conflict, an increasing academic effort has been undertaken to investigate the role of climatic and environmental determinants in fostering violence (Hsiang et al., 2013; Buhaug et al., 2014). While the empirical literature has failed to find a robust direct association between climatic variability and the likelihood of conflict, scholars have recently shifted their attention towards the possible indirect or conditional pathways leading from climate change to political instability (Koubi, 2018). One of the most widely-explored mechanism advanced by the literature on the climate-conflict nexus considers climate-induced shocks to agricultural production as a potential intermediate step (Miguel et al., 2004; Dell et al., 2014). Despite some qualitative analyses and case studies supporting the role of climatic-driven agricultural shocks in leading to higher risks of conflict, the findings of the empirical studies are rather controversial, featuring still limited evidence of a casual chain linking climate change, crop production, and conflict (Bollfrass and Shaver, 2015).

In the present analysis, we argue that the lack of robust empirical findings is not neces-

sarily driven by a non-existing relationship, but rather to a failure of the past literature to account for relative differences in crop yields both at the country level and between groups. Climate change is likely to alter the quantity and allocation of water and arable land, with resources increasing in some regions of the world and decreasing in others (IPCC, 2014), and thus raise the inequality in crop production. We therefore hypothesize that the destabilizing effect of climatic anomalies arises in those settings where the distribution of agricultural output is particularly unequal, leading to relative deprivation rather than mere scarcity. In this perspective, our empirical approach enables us to go beyond the conventional absolute measurement of crop production and explicitly focus on relative differences in agricultural yield. In doing so, we especially build on the academic debate around vertical and horizontal inequality, so as to investigate the agricultural dimension of inequality at different levels - both within countries and among identity-based groups. Originated from Gurr's theory of relative deprivation (Gurr, 1970), horizontal inequality (HI henceforth) is defined as the inequality in economic, social, political or cultural dimensions between identity-based groups (Stewart, 2008b). The main idea behind HI is that relative differences in socio-economic, political or cultural conditions between groups offer more insight in explaining conflict than the mere differences across individuals who do not share any precisely-defined identity, as for example in the case of socio-economic classes (Stewart, 2008a).

Here, we test the hypothesis that not only the inequality in socio-economic, cultural and political factors but also an unequal distribution of crop production, which is strongly related to food security, is positively related to the probability of intra-state conflict. We

hypothesize that higher levels of inequality in crop production both at the country level and among identity-based groups, further exacerbated by climatic fluctuations, increase the probability of intra-state conflict. To test this hypothesis, we rely on geo-referenced data on conflict relative to the period 1981-2017, satellite data on temperature and precipitation anomalies from the long-term mean, grid-cell information on socio-economic development, and high-resolution annual gridded data on the local yield of four main crops. Taking advantage of the spatially disaggregated, time-variant information available, we aggregate the gridded yields of four main crops to obtain an estimate of crop yields for each country, year and identity-based group. Our empirical approach enables us to compute a set of indicators measuring inequality in crop production at both the country and group level.

This study contributes to the literature in many regards. First, we introduce a more flexible definition of agricultural production as predictor of conflict, which is not a priori concerned with scarcity or abundance as absolute quantities, but follows the spirit of Selby and Hoffman's notion of "relative differences" (Selby and Hoffman, 2014). As climate change impacts on water and fertile land may vary across regions (Jiménez Cisneros et al., 2014), inequality in crop production may arise as a consequence of erratic climatic patterns and hence increase the relative agricultural deprivation, which in turn creates or exacerbates pre-existing societal grievances that may translate into violent actions. Therefore, we hypothesize that the link between climatic change and conflict may go through inequality in agricultural production, rather than the mere quantity of crop yields. Second, we advance the literature on inequality and conflict by examining the environmental dimension of inequality, which

has so far been neglected. Third, we investigate the inequality in crop production both at the country level ("vertical" inequality) and between identity-based groups within national units ("horizontal" inequality), in a way to assess the response of groups to different distributive patterns of agricultural production. Finally, we conduct a global analysis to avoid risks of a "streetlight effect" or an "African bias" threatening the results of conflict analyses (Adams et al., 2018).

The remainder of the paper is organized as follows: first, we present the main pathways that can lead from climate-induced agricultural shocks to conflict; next, we illustrate the data and methods used to build the set of inequality indicators; the final section presents the results and concludes.

2 Crop and Conflict

After a failure to find a robust direct correlation between climatic variability and conflict risk, the empirical literature on climate and conflict has recently shifted focus to explore the indirect or conditional pathways that can potentially transmit climatic fluctuations to increased likelihood of violence (Koubi, 2018). Among the pathways that have been hypothesized to lead from climate change to instability, one of the most widely-accepted mechanisms features climate-driven shocks to agricultural and food production (Wischnath and Buhaug, 2014) and advances that climatic variability will have a higher effect in rural and agriculturally-dependent areas, as temperature and precipitation anomalies are tight to crop growing season (Von Uexkull et al., 2016; Harari and La Ferrara, 2018).

Climate change is expected to decrease agricultural output, which in turn will have a

particularly destabilizing effect in developing countries, whereby governments may lack resources to implement adequate adaptation measures (Rosenzweig and Perry, 1994; Morton, 2007; Lobell et al., 2008; Mendelsohn, 2008). In turn, the negative impacts of climatic variations on agriculture may cause crop failures, reduce income from food production, deteriorate the local livelihood and sow the seed of conflict via different mechanisms: (i) climate-driven crop scarcity may generate or exacerbate pre-existing grievances (Malthus, 1798); (ii) the relative deprivation felt by destitute farmers may deepen inequalities and social fragmentation (Jones et al., 2017), and reduce the perceived opportunity costs of violence (Brinkman and Hendrix, 2011); (iii) exogenous shocks to crop yields may destabilize political settings by increasing the volatility of food prices and making developing countries more dependent on food imports (Hendrix and Haggard, 2015); and (iv) the failure to satisfy the most basic needs may push groups' outmigration towards urban areas (Bollfrass and Shaver, 2015).

On the contrary, as conflict is costly, the vulnerability resulting from resource deprivation can decrease both individuals' willingness and opportunity to join rebel groups (Salehyan and Hendrix, 2014). Along the opposite direction, the wealth of agricultural resources may also trigger looting strategies that result in higher levels of violence, as combatants must rely on local agricultural resources for sustenance (Koren and Bagozzi, 2017).

Yet, to date the evidence of a robust link between climate, agricultural production shocks and conflict is debated in the empirical literature. Harari and La Ferrara (2018) find evidence that crop failures during the growing season drive the local relationship between climate conditions and conflict episodes, while climatic shocks occurring outside the growing season

have no impact on conflict dynamics. Similarly, Von Uexkull (2014) shows the impact of drought on conflict to be conditional on agricultural output and demonstrates that areas with rainfed cultivations are more likely to experience conflict when hit by droughts.

In contrast, Bollfrass and Shaver (2015) cast doubts on the alleged indirect channel at the subnational level, showing that the link between temperature variation and conflict is significantly positive for both regions that experience fluctuations in agricultural output and those which do not exhibit any variation. Also, Buhaug et al. (2015), which focuses on food production as a key predictor of conflict incidence, finds no support to the claim that decreased agricultural production induced by climatic variability spurs conflict in Sub-Saharan Africa. Likewise, other studies fail to find evidence that rural or agriculturally-dependent societies are more at risk of conflict (Hendrix and Glaser, 2007; Bell and Keys, 2016).

In the present analysis, we argue that the ambiguity in empirical findings may be partly due the tendency of the existing literature to define agricultural production in absolute terms, while much less attention is paid to the relative differences in crop yields, either at the country level ("vertical" inequality) or among identity-based groups ("horizontal" inequality). Unequal endowments in crop yields, rather than the absolute stock of crops, may represent a significant correlate to conflict onset, whereby the relative agricultural deprivation may create or exacerbate pre-existing grievances and translate into effective action. As far as we are concerned with the relative differences in agricultural production, our empirical approach builds on the academic debate around the notions of vertical and horizontal inequality. The

definition of horizontal inequality has emerged in the conflict scholarship as a reaction to previous empirical analyses that dismissed the role of grievances in general, and inequality in particular, in triggering conflict (Fearon and Laitin, 2003; Collier and Hoeffler, 2004). In opposition to theorists of opportunity and greed, scholars have argued that the failure to find a link between inequality and conflict was not due to a true lack of evidence, but rather to a wrong conceptualization and operationalization of the independent variable (Cederman et al., 2013; Ostby et al., 2011; Ostby, 2013). While previous analyses focused on the so-called vertical inequality, or the inequality between individuals within the same group - typically, income inequality - this new wave of studies, building on Gurr's theory of relative deprivation (1970), highlights the role of differences between groups in explaining conflicts (Stewart, 2008b; Buhaug et al., 2011; Ostby et al., 2011).

In the original development of the HI argument, Stewart (2008b) advanced the hypothesis that the control over natural resources either directly or via the state, is a considerable source of group competition, particularly in resource-rich countries. However, the dimension of inequality related to agricultural resources have never, to the best of our knowledge, been tested as a predictor of conflict. Some studies did focus on land inequality as a potential predictor of insurgency, but they only concentrated on the maldistribution of arable land and were confined to a standard definition of vertical inequality (Russett, 1964; Muller and Seligson, 1987). Land maldistribution is not necessarily a good proxy of crop production; indeed, higher agricultural outputs can be achieved without necessarily expanding the amount of land per capita, through enhanced agronomic practices, improved crop varieties, and other

technological innovations (Tilman et al., 2011).

The present study tries to avoid the limitations of the previous literature and focuses on the inequality in crop production both at the country level (vertical inequality) and between identity-based groups (horizontal inequality). We may schematically think of two possible pathways through which inequality in crop production can induce conflict: first and more simply, it can act across individual units (grid-cells in our study) within countries; second, it can operate between individuals belonging to different groups, and thus intertwine with pre-existing identity based dynamics and cleavages. Borrowing from the literature on economic or income inequality, we call the first type of mechanism "vertical" inequality in crop production, as it compares individual units without necessarily sorting them into categories or taking into account the characteristics of each category (Cederman et al., 2011); we define the second pathway as "horizontal" inequality in crop production, whereby it considers the internal diversity of distinct identity-based groups. The difference between the two types of inequality is far beyond a mere technicism, as horizontal inequality accounts for an identity-related dimension which is absent in the vertical one.

Crop inequality does not necessarily need to overlap with identitarian cleavages to be inducive of conflict, as the relative agricultural deprivation borne by different locations within countries may trigger a sense of frustration that pushes individuals to mobilize to advance their grievances. However, the second type of inequality in crop yields (horizontal) is likely to be especially destabilizing, whereby it overlaps with pre-existing identity-based cleavages, and can hence exacerbate grievances operating along socio-economic, ethnic or cultural lines

(Figure 1).

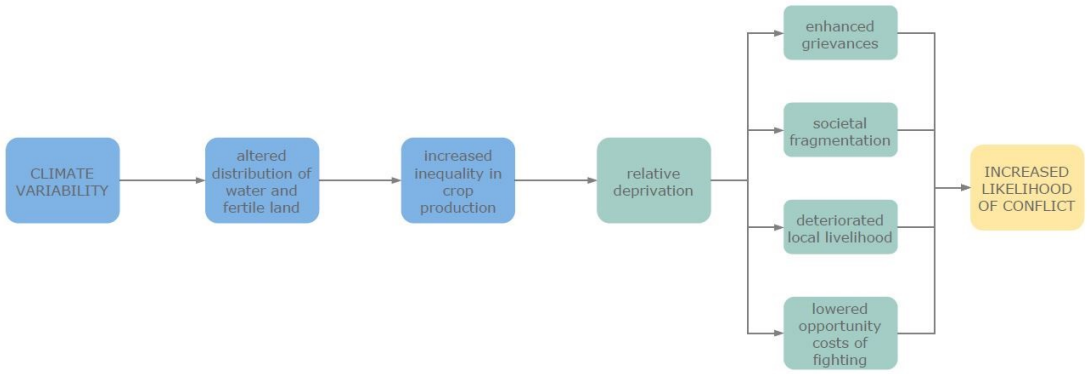

Figure 1: Main pathways leading from climate variability to increased likelihood of conflict through inequality in crop production. For simplicity, the links are kept at a minimum.

As the geographic categorization is sensitive to individuals' self-determination and often overlaps with ethnic boundaries in ethnically heterogeneous societies, our analysis concentrates also on the inequality among regional groups, rather than being limited to ethnic groups (discussed in detail in Section 2.1).

The impacts of climatic change on agricultural production are likely to further exacerbate the inequality across spatial locations, as the availability of crop yields are expected to increase in some regions of the world while decreasing in others (Zhang and Cai, 2011). This effect is mainly due to reduced freshwater availability, temperature increase, as well as higher frequency and intensity of extreme events, although the influences of climate on crop yield are likely to be multiple, complex and possibly indirect. For instance, climate variability is predicted to alter the timing and magnitude of runoff and soil moisture, affect lake levels, impact water quality (Gleick, 1993), and in general result in limited access to freshwater resources, due to the increasing costs borne by the water supply sector in dealing

with erratic rainfall patterns and unpredictable water levels (Porter et al., 2014).

The detrimental effect of climate variability on crop production is expected to vary both in time and space, and will be especially detrimental in developing countries (Ramankutty et al., 2002; Arnell, 2004; Scheffran and Battaglini, 2011), whereby vulnerable populations characterized by low levels of socio-economic developments may be less resilient to the destabilizing shock of environmental and climatic hardships. Solid scientific evidence holds that poverty and low levels of development are detrimental for countries' peace and stability (Collier and Hoffler, 1998; Hegre, 2001; Fearon and Laitin, 2003; Miguel et al., 2004; Collier and Rohner, 2008). So strong is the consensus in the conflict scholarship around the link between socio-economic conditions and violence that conflict has been referred to as "development in reverse" (Collier, 2003). Likewise, a number of studies demonstrate that higher the level of socio-economic development, the lower the risk of conflict (Hegre and Sambanis, 2006; Blattman and Miguel, 2010; Hegre et al., 2016). Hence, groups characterized by better socio-economic conditions can be more resilient to external shocks than deprived or marginalized societies, whereby the effect of inequality in crop production will be moderated by the safety valve of development. By contrast, poor locations characterized by low development rated will be more vulnerable to environmental hardships and external shocks (Adger, 2000). Consistently, in our analysis we do control for the role of socio-economic conditions (GDP, population size) in affecting the risk of civil conflict.

The line of reasoning illustrated in this section enables us to formulate the following hypotheses:

Hypothesis 1a. Higher levels of inequality in crop production at the country level ("vertical inequality") are associated with a higher probability of intra-state conflict onset.

Hypothesis 1b. Higher levels of inequality in crop production between identity-based groups ("horizontal inequality") are associated with a higher probability of intra-state conflict onset.

Hypothesis 2. The effect of climate anomalies on civil conflict onset is conditional upon the level of inequality in agricultural output.

To test our main hypotheses, we adopt both a country-level and a group-level unit of analysis, which enables us to focus on relative differences among regional/ethnic groups. The next paragraph illustrates the choice of the scale of analysis in more detail.

2.1 Inequality in crop production and conflict

The literature on intra-state conflict onset tends to focus either on individuals at the societal level or at the country-level, while neglecting both the sub-state and the structures operating between these two levels (Buhaug et al., 2011). The factors that may explain conflict, however vary spatially within countries (Buhaug and Rod, 2006), and ignoring the sub-state spatial variation of potential explanatory factors prevents a nuanced testing of the relationship between geographically clustered resource allocation and the likelihood of conflict at the sub-national level. We overcome the problems associated with national studies by using disaggregated, geo-referenced data on crop production covering years 1981-2014 (Iizumi et al., 2018) and aggregating them at the regional or ethnic group level. These data allow us to investigate whether inequality in crop production across geographically defined groups is

associated with a higher probability of civil conflict.

Theoretically, many different types of social cleavages, including class-based ones, can be linked to conflict. However, although they recently began attracting attention in the policy and academic field, spatial and regional inequalities and their consequences have so far only limitedly been documented and analyzed (Ostby et al., 2009). The conflict scholarship has indeed devoted most of the attention to ethnic distinctions, since they are ascriptive and highly visible per sé (Horowitz, 1985). This has determined a tendency of the empirical literature of measuring inequality along a single dimension, i.e. ethnicity, neglecting other plausible dynamics of polarization among exogenously given groups, such as urban/rural or regional groups (Ostby, 2008). Nevertheless, around half of the civil conflicts after the end of the Cold War has been classified as "ethnic" (Fearon and Laitin, 2003; Esteban et al., 2012), meaning that "all conflict is not ethnic" (Esteban et al., 2012, p. 1321), or in other words, not all conflicts are ethnic conflicts and not all ethnically-divided societies are violent (Fearon and Laitin, 2003; Stewart, 2008b). There is therefore a need for research on conflict to go beyond the usual notions of primordial antagonisms that are generally associated with ethnicity (Cederman et al., 2011).

The present study aims precisely at investigating the effect of inequality in crop production among regional groups, other than purely ethnic entities. This enables us to explore both ethnic and non-ethnic conflicts, to assess whether they may be linked to the distribution of crops at the sub-national level.

Regional borders reflect natural and geographical, as well as historical and political differ-

ences (Ostby et al., 2009), and may thus be more relevant to agricultural production than the pure ethnic stratigraphy. As insightfully argued by political geographers, borders represent institutions, whose main essence is to separate the "self" from the "other" (Paasi, 1998; Newman, 2003). In this view, territory constitutes the main dimension of identity, and borders mirror existing differences and socially construed characteristics which are associated with the historical imaginary of the homeland (Kaplan, 1999). Identity is, as such, broader than ethnicity, encapsulating notions of social categories and laying on the essential definition of the spatial and temporal context in which it applies (Fearon, 1999). Even though the original definition of Horizontal Inequality characterizes groups as identity-based (Stewart, 2008b), scholars of the HI-conflict nexus have so far neglected the multifaceted, complex nature of identity to simplistically assume that group identity and ethnicity coincide. We argue that regional groups, as demarcated by sub-state boundaries, represent a comprehensive and exhaustive instance of identity-based groups and we investigate the effect of inequality in crop production, not only among ethnic but also regional groups, on the likelihood of intra-state conflict onset. The next section illustrates in detail the data and methods we apply to test our main argument.

3 Research design

3.1 Indicator of Inequality in Crop Production

To our knowledge, no other quantitative global study of intra-state conflict has examined the relationship between inequality in crop production and conflict. Studies have extensively examined the destabilizing effect of economic inequalities across identity-based groups (Gurr,

1970; Collier and Hoeffler, 2004; Cramer, 2003; Ostby, 2008), but much less attention has been paid to the conflict potential of agrarian inequality (Russett, 1964; Muller and Seligson, 1987). Yet, studies of inequality in agrarian resources suffer from two main flaws: first, they are confined to land tenure maldistribution, while totally neglecting the differences in land production; second, despite framing their theories in regional terms, they only focus on vertical inequality and use the national state as their unit of analysis (Paige, 1978). Nevertheless, as sharing a common identity is essential for a group to mobilize (Tilly, 1978; Gurr, 2000), agrarian inequality may be relevant not only for different classes of individuals within the same group, but also across groups.

This study tries to avoid some of the limitations of the previous literature by building a set of empirically based indicators of inequality in crop production, based on geo-referenced data. We compute two main measures of inequality: i) the simple Gini coefficient as a measure of inequality in crop production at the country level ("vertical inequality"); ii) the Gini coefficient across identity-based groups - regional and ethnic - as a measure of "horizontal" inequality in crop production.

The plain *vanilla* Gini coefficient is computed on the basis of the value of crop production for each grid-cell, nested within national countries, and thus measures the variance in yields across grid-cells as individual units. The between-groups Gini index for a country's population consisting of n groups with values of crop production y_i is defined as:

$$G = \frac{1}{n}(n+1-2)\frac{\sum_{i=1}^{n}(n+1-i)y_i}{\sum_{i=1}^{n} y_i}. \tag{1}$$

where group-level values of crop production y_i are indexed in non-decreasing order such that $y_i \leq y_{i+1}$. The between-groups Gini index captures differences in the mean crop production across groups (Alesina et al., 2016); in our case, this relies on the assumption that each group has the same access to agrarian products, or in other words, that the benefits deriving directly or indirectly from crop production are gained locally by each group, rather than being appropriated by the central government. Although this is quite a strong assumption, the lack of data on property rights and degrees of access to agricultural benefits by each group prevents us from conducting a more detailed analysis. Moreover, while the issue of rents' redistribution is surely worth-considering in the case of profitable resources like oil and minerals, it is less relevant to agricultural production, whose gains are likely to be accessed homogeneously by groups - especially in the least developed areas, not yet characterized by the oligopolist structure and the dominance of few big crop producers that is more frequent in developed areas.

For our purpose, we group individuals by "non-agrarian" characteristics, *i.e.,* features which are different than the variable for which the inequality is being calculated: first, by regional identity, and second, by ethnicity. To compute the mean value of crop production for each n group within country's population, we spatially overlay groups' geo-referenced polygons with gridded data on crop production (Iizumi et al., 2018) and aggregate the cell-level crop yield to obtain the mean value of agricultural production by regional/ethnic group. Information on regional groups' settlements and borders corresponds to the second

administrative level of GADM global data (GADM, 2018); ethnic groups are defined by using the spatial polygons provided by the 2019 version of the Geo-EPR dataset (Vogt et al., 2015).

For each spatial group, we construct two basic measures of the Gini index: the first one uses the information on crop production from all grid-cells; the second measure is limited to rural grid-cells and excludes urban areas from the sample.[1] Indeed, when measuring the inequality in crop production across groups, a potential issue may arise concerning the different crop yield levels that characterize urban vs. non-urban areas: urban areas will have low or null production of crops and could therefore erroneously be interpreted as "poor" in agricultural value compared to rural locations. As our main goal is to observe the effect of an unequal distribution of crop among regional or ethnic groups, including urban areas in the analysis may potentially influence the construction of the Gini index, as it would consider urban cells equal to rural locations which have a low or null level of crop production.

Figures 2 and 2 visualize our measures of inequality for both the full sample and only rural cells. Even a quick glance at the maps enables us to realize that two orders of factors drive the difference in the values of the Gini coefficient: a) the way of measuring inequality, either vertically (standard Gini index) or horizontally (between-groups Gini index); b) the definition of groups according to different identity-based characteristics (regional, ethnic). By contrast, the distribution of inequality in crop production is not considerably affected by the exclusion of urban areas from the sample, as the values of the Gini index tend

[1]Rural cells are defined as those whose land surface is not covered by any artificial/urban structure, according to Globcover categorization of land-use areas (Bontemps et al., 2009). Urban cells are any cells for which the extension of artificial/urban area is greater than zero.

to follow the same pattern whether urban areas are included or not. While inequality in crop production between ethnic groups is generally low, inequality between regional groups displays the highest variation across countries. This is consistent with our intuition that the distribution of natural resources and crop productivity follows the regional boundaries rather than the ethnic ones. The vertical inequality in crop production within countries seem to be homogeneously high all over the world and especially in developed countries. At first, we may be surprised to observe that wealthy countries such as the United States have higher levels of inequality in crop production than poor countries.

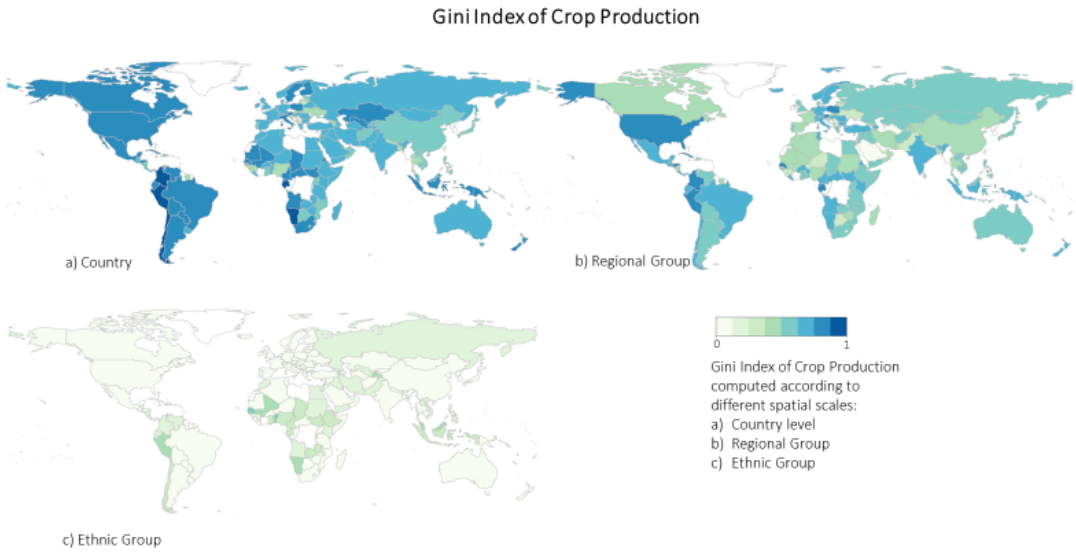

Figure 2: Gini index of crop production in urban and rural areas. a) Simple Gini Coefficient, computed for grid-cells nested within countries, as a measure of vertical inequality; b-c) Between-groups Gini Coefficient computed for regional and ethnic groups nested within countries, as a measure of horizontal inequality.

However, this is coherent with the technological advances in cultivation techniques and the switch to intensive farming that characterize the latest stages of economic development,

leading to an oligopolistic concentration of crop production in the hands of few producers (Sexton et al., 2007).

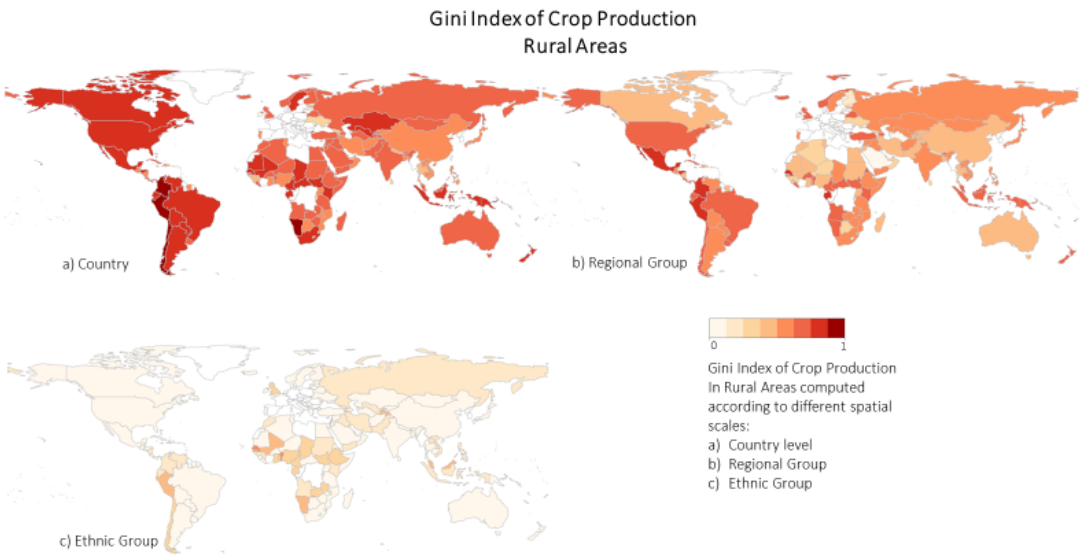

Figure 3: Gini index of crop production in rural areas. a) Simple Gini Coefficient, computed for grid-cells nested within countries, as a measure of vertical inequality; b-c) Between-groups Gini Coefficient computed for regional and ethnic groups nested within countries, as a measure of horizontal inequality.

Economic inequality has been shown to observe a similar trend, with developed countries featuring higher levels of income inequality than nations at lower levels of development, according to a U-shape relationship (Anand and Kanbur, 1993; Stiglitz, 2013).

A potential issue in measuring inequality between groups is represented by the heterogeneous dimensionality of groups within countries; as the number of individuals in each group is likely to differ, an unweighted index would assign the same weight to the relative position of small and large groups (Stewart, 2008a). To account for differences in size across groups, in our robustness tests we construct a population-weighted index, by replacing the standard

group-level mean with a population-weighted average of crop production for each group. Regression results relative to the population-weighted index are available upon requests.

3.2 Empirical model and Data sources

The effect of crop inequality on the probability of conflict onset is estimated by means of the following empirical model:

$$Y_{i,t} = \alpha + \beta Gini_{i,t} + \gamma A_{i,t} + \mu(Gini_{i,t} \times A_{i,t}) + \delta X_{i,t} + \theta E_{i,t} + \epsilon_{i,t} \tag{2}$$

where $Gini_{i,t}$ is the Gini coefficient of crop production inequality (as described in the previous section) in country i in year t; "A" is the vector of climatic anomalies (precipitation and temperature anomaly), "X" contains controls for a country's level of development, population size and other country-specific variables; "E" is the vector of ethnic diversity and ethnic politics variables, and "C" contains a complete set of geographical and time controls.

The main dependent variable "Y" is the intra-state conflict as defined by the Uppsala Department of Peace and Conflict Research (UCDP, 2019). We consider all conflict types over the period 1981 - 2017, occurring within the borders of a state, which involve the use of the armed force by either the government or a formally organized group, and results in at least 25 battle-related deaths in a year. We code the dependent variable (onset of intra-state conflict) as a binary variable coded as "1" for every year in which a conflict breaks out in a given location and "0" otherwise. Data on conflict are drawn from the recently released UCDP-GED dataset, a geo-referenced gridded version of UCDP data (Sundberg

and Melander, 2013). As the UCDP-GED data only covers the years after 1989, we integrate it with the PRIO-UCDP dataset for the period 1981-1989, as reported in the PRIO-GRID structure (Tollefsen et al., 2012). The intra-state conflict category broadly includes three types of violence: (i) state-based conflict, which involves at least a government of a state as an active part; (ii) non-state based conflict, which is the use of the armed force between two organized groups such as rebel groups; and (iii) one-sided violence, which is defined as the use of the armed force by either the government of a state or an organized group, against the civilians. As there is no reasonable motivation to think that unequal levels of crop production have a greater effect on one specific type of intra-state conflict, our analysis takes all the three subcategories of intra-state conflicts into consideration. Second, we test for the effect of inequality in agricultural production on non-state conflicts, such as those between rebel groups or non-governmental actors. Further, we also consider the onset of ethnic civil conflicts, defined as a binary variable that takes the value of "1" in the first year of an armed ethnic conflict and "0" otherwise. A conflict episode is defined as ethnic whenever rebel organizations pursue ethno-nationalist aims and recruit along ethnic lines in order to achieve ethno-national self-determination, a more favorable ethnic balance of power in government, ethno-regional autonomy, the end of ethnic and racial discrimination, or a more balanced division of resources along ethnic lines within society. We utilize country-level data on ethnic conflicts from the latest version of EPR dataset (Voigt et al., 2015).

To build the crop production inequality indicators (Gini), we rely on information on the annual production of four main crop types (maize, wheat, soybean, rice) in a given

grid cell, expressed in tonnes per hectare (t/ha). Global gridded data at 0.5 deg resolution ($\sim 55km \times 55km$ at the equator) are drawn from Iizumi et al. (2018) which improves upon the spatial resolution of the previous versions of their dataset (Iizumi et al., 2014; Iizumi and Ramankutty, 2016) . Although the coverage of the data, which includes only four crop types, may represent a limitation to the analysis, we believe this is an acceptable price to pay for building a time-variant dataset, which is necessary to investigate the conditional effect of climatic variability on conflict onset.

To test the hypothesis that climatic variability would increase the impact of crop inequality on the risk of conflict, we interact the measure of inequality in crop production with data on temperature and precipitation anomalies. Specifically, we define climatic anomalies as the standardized deviation of annual temperature and precipitation from their long-term climatologic mean, conventionally defined as the mean over a period of almost thirty years (WMO, 2017). Gridded climatologic data on temperature and precipitation are retrieved from monthly meteorological statistics developed by the University of East Anglia Climatic Research Unit (University of East Anglia Climatic Research Unit, 2017). Data are provided on high-resolution ($0.5 \times 0.5 degree$) grid cells and aggregated at the country level.

Finally, we include a number of control variables to account for socio-economic contextual factors that can increase or decrease the odds of violence. A full list of the controls and descriptive statistics are presented in Tables 10 and 11. Our main covariates are standard in conflict research: GDP per capita, population size, characteristics of the political system, educational attainment of the population, oil rents, incidence of international trade

in domestic GDP, and geophysical characteristics (mountainous terrain).

We use Gross Domestic Product per capita (GDP per capita) as a proxy for socio-economic development (Ray and Esteban, 2017). Data on GDP per capita are drawn from a gridded annual database for the whole world at 0.5° resolution for the 25-year period of 1990-2015 (Kummu et al., 2018). Data for the missing years are linearly interpolated. In order to cope with endogeneity, GDP per capita and population size are lagged by one year.

In order to characterize the political systems, we use the 2019 version of V-Dem data set (Coppedge et al., 2019) based on an expert evaluation of the countries' level of democracy. V-Dem provides a comprehensive definition of regime types, which takes into account the complexity of the concept of democracy as a system of rule that goes beyond the simple presence of elections. Specifically, we code a variable for autocracy, corresponding to regime types 0 and 1 in V-Dem Regimes of the World measure.

As for additional country characteristics, we consider the share of the territory covered by water, terrain roughness and the length of the drought season as a percentage of the year. Geophysical or structural characteristics such as terrain roughness, which enhance rebels' chances of hiding, may indeed favor rebellions and insurgencies (Fearon and Laitin, 2003), while drought may increase the probability of conflict by negatively affecting the agricultural production (Harari and La Ferrara, 2018). These controls are obtained by aggregating the cell-level information to obtain the mean value of water and mountain coverage and the mean duration of the drought seasons by country and year. The drought variable is lagged by one year, as the detrimental effect of droughts on agricultural production are likely to

be perceived in the long-term rather than immediately. Finally, since countries which rely heavily on the agricultural sector will be more sensitive to changes in crop yields and therefore more subject to exogenous shocks to the agricultural production, we controlled for the degree of agricultural dependence. As a crude proxy for agricultural dependence, we used data on the value added of agricultural production (World Bank, 2012).

As ethnic competition is likely to affect conflict risk (Cederman et al., 2013), we also include controls for the ethnic composition of the population, the degree of ethnic exclusion and the intensity of competition between different power groups defined along ethnic lines. Specifically, we control for the degree of ethnic polarization by means of the discrete polarization index developed by Montalvo and Reynal-Querol (2002, 2005); the degree of exclusion along ethnic lines is measured by the logged share of the population excluded from the central government; and the number of power sharing groups, represented by ethnic elites at the central government, proxies central power segmentation, i.e. the political competition between groups in the decision making process.

Finally, we consider a set of country-specific controls for the geographic region (macro-area), regional time trend, and the number of peace years since the outbreak of a previous conflict.

4 Results

The results of the regression analysis are presented in Tables 1 - 9. We consider three different classifications of conflicts: (i) an aggregate category containing all types of conflicts, (ii) conflicts that are considered purely ethnic in nature, and (iii) a separate category containing

only non-state conflicts. The Gini index is calculated by using the information on agricultural production at cell level for two alternative cell sub-samples: (i) all cells with no distinction between urban and rural areas (Models 1-3), and (ii) only rural areas (Models 4-6). In addition to the distinction between urban and rural areas, we also consider three alternative territorial (spatial) aggregations of agricultural productivity. We first calculate the Gini index at the country level (using the information at cell level with no additional aggregation) for each available year in our sample (1981 - 2017). This measure is used as an indicator of vertical inequality in crop production across countries and across time (Tables 1 - 3). Second, we aggregate the information on crop production at the regional group level and calculate the between group Gini index. This indicator aims at measuring the degree of (horizontal) inequality between distinct regional groups (Tables 4 - 6). Finally, we replicate the calculation of the Gini index using ethnic instead of regional marker to identify relevant social groups (Tables 7 - 9). The models step-wisely increase the set of predictors. We start by including the Gini index together with a full set of country-specific controls and geo-physical characteristics, as well as a complete battery of climatic variables. The models are than gradually enriched to include the interaction terms between the Gini index and proxies for climatic variability, as well as three additional time varying variables related to ethnic composition, ethnic exclusion and discrimination, and ethnic competition. In all model specifications we control for geographic macro-area and regional time trends, and correct for error correlation over time for a given country by calculating cluster - robust standard errors.

4.1 Vertical and Horizontal Inequality in Crop Production

Results of the empirical analysis confirm our first hypothesis (*Hypothesis 1a* and *1b*). Indeed, we find that both vertical and horizontal inequality in crop production are associated with a higher probability of conflict onset. This effect is particularly pronounced for rural areas and for a wide definition of conflict onset (all types of conflicts). A marginal increase in inequality of agricultural production at the country level (considering both urban and rural areas), holding the other variables at their mean, leads to a 18% higher probability of civil conflict outbreak, and the effect is 10 percentage points higher when we consider only rural areas (Table 1). This is consistent with the previous literature's findings that rural regions tend to be more vulnerable to environmental hardships (Von Uexkull et al., 2016): an exogenous stress to agriculturally dependent populations would hit them more heavily than urban groups, who can rely on a diversified set of resources. The effect of agricultural inequality is significantly reduced for ethnic conflict onset (7%) and becomes insignificant in the case of non-state conflicts (Tables 2 and 3).

The inequality in crop production, hence, has a destabilizing impact on societies: unlike pure economic inequality, which is not significantly associated to violence (Collier and Hoeffler, 2004), crop maldistribution across individual units (grid-cells) within the same country is conducive of conflict. In other words, the conflict potential of agricultural inequality unfolds even when there is no clear-cut idea of group identification, and maldistribution in agricultural output is per sè a destabilizing factor, without necessarily overlapping with ethnic or other types of identity-based dynamics to trigger frustration and exacerbate grievances.

However, as the majority of civil conflicts are organized group conflicts (Stewart, 2008c, p. 11), the mobilization of people is easier when it targets groups with strongly defined, shared identities, rather than individuals. The conflict potential of inequality in crop production may be important as well (or even more pronounced) in the case of precisely defined identities between groups, as it can exacerbate societal grievances, induce societal fragmentation, lower the perceived opportunity costs of fighting and hurt the local livelihood to the point that it translates into violent action. The coefficients on Gini index in Table 4 confirm that inequality in crop production between regional groups represents an important trigger to civil conflict onset in rural areas, although the magnitude of the effects is not significantly different from the one computed at the country level in Table 1 and loses significance when urban areas are included in the analysis. The effect of crop inequality on non-state conflict outbreak is again significant only for rural areas, with a marginal impact equal to 18%.

The importance of regional or geographic distribution in crop production over ethnic differences is corroborated by Models 1-6 presented in Tables 7 - 9, where we find that inequality in agricultural productions between ethnic groups is not significantly correlated to conflict (at least across rural areas), with the exception of ethnic conflicts, where it remains significantly different from zero in all model specifications. The results suggest, moreover, that among ethnic groups, socio-political and identity-based characteristics, akin ethnic competition and discrimination, are important predictors of conflict outbreak. Not surprisingly, these effects are more pronounced in rural areas.

The empirical evidence presented so far suggests that regional and geographic borders

are a more comprehensive measure of identity and may be more relevant to crop production than ethnic differences, but also signals that the spatial definition of groups, along regional or ethnic borders, is of foremost importance in explaining violence. In other words, different types of groups appear to react differently to socio-political, environmental and climatic stresses. Conditions that are inducive of conflict for a particular identity-based group may not be so for another group type.

As for the other common determinants of conflict onset, the level of GDP per capita is not significantly correlated with the probability of conflict outbreak; this is not surprising after all, since part of the effect of GDP is absorbed by agricultural production. Consistently with a well-established conflict scholarship, we find that the size of the population and the number of peace years since the outbreak of the previous conflict are significantly related to conflict onset. In contrast to Fearon and Laitin's (2003) insurgency model, oil production per capita and mountainous terrain receive limited support here, with the exception of ethnic conflict onsets where oil production is significant at the 0.01 level. Finally, although the coefficients associated to autocracy have the expected sign, they do not reach significance at the 0.05 level.

4.2 Climatic Variation, Inequality and Conflict

The empirical evidence on the role of climatic variability in increasing the probability of conflict outbreak is mixed. Drought episodes are significantly correlated with the outbreak of civil conflicts, while are generally not significant when we consider only ethnic and non-state conflict onset. Interestingly, the results suggest that climatic variables have a very

modest, if any, influence on the probability of conflict onset, and that the effect is heavily dependent on the type of conflict chosen as a dependent variable, as well as the way to operationalize the predictor. The direct impact of temperature and precipitation means on civil conflict is generally null and insignificant. The average temperature appears to be a significant correlate of civil conflict onset only for one specification of the model (Model 6, Table 1), when we compute the Gini coefficient at the country level (vertical inequality), but the effect is not robust to alternative specifications.

Temperature and rainfall means have a limited effect on non-state conflict; the results show that lower levels of rainfall slightly increase the likelihood of non-state violent episodes, while increases in the average temperature are associated with a higher probability of non-state conflict onset (Tables 3, 6, 9), although the effect is very modest. Non-state conflicts identify less structured types of violence than civil conflict, as the fighting occurs between rebel groups or actors without the involvement of the government. Our results may therefore suggest that climatic variables are more likely to trigger lower-intensity and less-organized form of violence, or to exert an influence over conflicts that are still at the embryonic form of riot or protest. By contrast, more structured form of conflicts, involving the national authority, do not seem to be affected by climate. This hypothesis is consistent with recent case studies on the Arab Spring and the Syrian conflict (Femia and Werrell., 2013; Gleick, 2014).

Temperature and precipitation anomalies or deviations from their long-term mean, on the other hand, do not have a significant impact on the likelihood of conflicts at the country

level and rainfall anomalies are shown to have only a limited effect at the regional-group level (Tables 4 and 5). The impact of climate anomalies remains null and insignificant even when interacted with the Gini Index of crop production, with the exclusion of ethnic conflict, when the analysis is performed at the regional level (Table 5). This non-effect of climate anomalies conditional to agricultural output is consistent with previous findings (Wischnath and Buhaug, 2014; Bollfrass and Shaver, 2015).

The failure to find any impact of climatic anomalies on conflict onset does not necessarily mean that climatic change has no destabilizing effect at all, but may indicate that socio-economic systems are to some extent able to adapt to the consequences of climatic variability, especially since the impact of variations in temperature and precipitation unfolds at a rather slowly pace. However, as we showed earlier, more abrupt climatic shocks like droughts are proven to significantly increase the probability of conflict across all model specifications considering the aggregate category of civil conflict onsets.

5 Conclusions

This study investigates whether relative differences in crop production, both within countries ("vertical inequality") and between groups ("horizontal inequality"), affect the probability of conflict onset and aims to understand the role of climatic changes in conditioning this relationship. We find that the inequality in agricultural production both at the country level and between regional groups is associated with a higher probability of conflict onset, while a maldistribution of crop yields between ethnic groups does not entail violence. Our undertake of the results is that the sense of frustration determined by a maldistribution of

crop production is conducive of conflict without necessarily overlapping with ethnic or other strong identity-based cleavages, whereby it operates also across grid-cells without precisely defined shared identity. However, the effect of inequality in agricultural output is slightly more destabilizing when it intertwines with pre-existing regional borders and rural areas are especially vulnerable to unequal distributive patterns of crop yields. The spatial definition of identity-based groups is indeed a meaningful discriminatory factor in explaining the relationship between inequality in crop production and conflict risk; regional and ethnic groups in fact seem to react differently to exogenous stresses and ethnic discrimination is confirmed to be a much better predictor than environmental and climatic characteristics in explaining conflict outbreak at the ethnic group level.

Although climatic variability is likely to alter the distribution of arable land and water and thus affect crop productivity, we do not find any support to the claim that temperature and precipitation anomalies affect the relationship between inequality in agricultural production and civil conflict onset. Temperature and precipitation play a limited role in increasing the probability of non-state violent episodes, indicating that the effect of climate is dependent on the type of conflict under investigation, as well as the operationalization of the climatic variables. By contrast, droughts are shown to be a significant correlate of civil conflict outbreak, and the effect remains significant throughout all the model specifications. This may indicate that populations are resilient enough to cope with the relatively slow-pace climatic changes proxied by temperature and precipitation anomalies, but less so in the case of abrupt climatic shocks such as droughts.

Further research is therefore needed to better understand the impact of more rapid-onset climatic changes, such as floods or natural disasters, on the distribution of agricultural yields and, in turn, on conflict. Also, the analysis can be extended to test how socio-economic characteristics, such as the level of development, and institutional elements, such as governance and trust, may condition the relationship between inequality in crop production and conflict onset. More efficient institutional systems, able to guarantee a fair distribution of property rights and the enforcement of cooperative agreements to manage resources, can in fact mediate the negative consequences of climatic changes and attenuate tensions.

Finally, a limitation of the present study is represented by limited data availability; indeed, grid-cell level information on crop production, utilized to compute the agricultural inequality at the country level, may not entirely correspond to the agricultural output available to each household. To this end, the study of inequality in crop production will surely benefit from ongoing progresses in data collection, which will hopefully make available high-quality time-variant survey data on agricultural production at the household level, thus enabling to better investigate the impact of "vertical inequality" in cultivation on the probability of conflict.

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Regression Tables

Table 1: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **COUNTRY level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ALL**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.177**	0.178**	0.161**			
	(0.078)	(0.078)	(0.081)			
Gini (rural)				0.284***	0.285***	0.266**
				(0.102)	(0.103)	(0.105)
Drought (% prev. year)	0.306	0.304	0.275	0.435**	0.432**	0.402
	(0.175)	(0.175)	(0.174)	(0.216)	(0.216)	(0.215)
Precipitation (yearly mean, lagged)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.004	0.004	0.004	0.007	0.007	0.009**
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
Precipitation (deviation, lagged)	0.013**	-0.018	-0.020	0.017**	-0.003	-0.007
	(0.006)	(0.022)	(0.022)	(0.007)	(0.027)	(0.027)
Temperature (deviation, lagged)	-0.006	0.011	0.006	-0.009	0.008	0.002
	(0.005)	(0.022)	(0.021)	(0.006)	(0.025)	(0.025)
Gini (all)*Precipitation (dev.)		0.049	0.051			
		(0.032)	(0.032)			
Gini (all)*Temperature (dev.)		-0.027	-0.020			
		(0.034)	(0.034)			
Gini (rural)*Precipitation (dev.)					0.032	0.036
					(0.039)	(0.040)
Gini (rural)*Temperature (dev.)					-0.027	-0.018
					(0.039)	(0.039)
Agriculture (% Gdp)	0.001	0.001	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Water Coverage (% terr.)	-0.000	-0.000	0.000	-0.001	-0.001	-0.000
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Mountains (% terr.)	0.035	0.035	0.039	0.096	0.095	0.105
	(0.058)	(0.058)	(0.056)	(0.076)	(0.076)	(0.071)
Gdp per capita (log, lagged)	0.016	0.015	0.011	0.024	0.024	0.019
	(0.011)	(0.011)	(0.012)	(0.014)	(0.014)	(0.015)
Population (log, lagged)	0.138***	0.137***	0.119**	0.166***	0.166***	0.161**
	(0.048)	(0.048)	(0.052)	(0.061)	(0.061)	(0.066)
Education (log)	0.013	0.013	0.011	0.013	0.013	0.011
	(0.010)	(0.010)	(0.010)	(0.012)	(0.011)	(0.011)
Oil (log)	0.006	0.006	0.005	0.006	0.006	0.005
	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.006)
Trade (log)	0.010	0.010	0.012	0.005	0.006	0.007
	(0.012)	(0.012)	(0.012)	(0.013)	(0.014)	(0.013)
Autocracy	0.024	0.024	0.022	0.023	0.023	0.020
	(0.031)	(0.031)	(0.034)	(0.035)	(0.035)	(0.039)
Number of Peace Yrs.	-0.007***	-0.007***	-0.007***	-0.007***	-0.007***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.046			-0.080
			(0.069)			(0.090)
Excluded Pop. (log)			0.126**			0.135**
			(0.059)			(0.069)
Number Ethn. Grps. in Power			0.004			0.003
			(0.007)			(0.008)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	4330	4330	4330	3583	3583	3583

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (all types of conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 2: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **COUNTRY level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ETHNIC**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.050**	0.052**	0.042			
	(0.024)	(0.024)	(0.025)			
Gini (rural)				0.067**	0.070**	0.056
				(0.029)	(0.029)	(0.031)
Drought (% prev. year)	-0.059	-0.059	-0.059	-0.074	-0.071	-0.075
	(0.053)	(0.053)	(0.052)	(0.064)	(0.062)	(0.062)
Precipitation (yearly mean, lagged)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.000	0.000	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Precipitation (deviation, lagged)	0.000	-0.011	-0.012	0.001	-0.011	-0.012
	(0.002)	(0.010)	(0.010)	(0.003)	(0.011)	(0.011)
Temperature (deviation, lagged)	-0.001	-0.001	-0.002	-0.000	-0.004	-0.004
	(0.002)	(0.012)	(0.012)	(0.003)	(0.014)	(0.014)
Gini (all)*Precipitation (dev.)		0.018	0.018			
		(0.014)	(0.014)			
Gini (all)*Temperature (dev.)		0.001	0.002			
		(0.018)	(0.018)			
Gini (rural)*Precipitation (dev.)					0.018	0.019
					(0.016)	(0.016)
Gini (rural)*Temperature (dev.)					0.006	0.006
					(0.022)	(0.022)
Water Coverage (% terr.)	-0.000	-0.000	-0.000	-0.001	-0.001	-0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Mountains (% terr.)	0.008	0.009	0.009	0.014	0.014	0.016
	(0.011)	(0.011)	(0.010)	(0.015)	(0.015)	(0.015)
Gdp per capita (log, lagged)	0.002	0.002	0.000	0.003	0.003	0.001
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Agriculture (% Gdp)	0.001***	0.001***	0.001**	0.001**	0.001**	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Population (log, lagged)	0.031**	0.031**	0.023	0.032**	0.032**	0.026
	(0.013)	(0.013)	(0.013)	(0.015)	(0.015)	(0.016)
Education (log)	0.005**	0.005**	0.003	0.006**	0.006**	0.004
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)
Oil (log)	0.003***	0.003***	0.002**	0.003***	0.003***	0.003**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Trade (log)	-0.002	-0.002	-0.002	-0.003	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Autocracy	0.002	0.002	0.006	-0.001	-0.001	0.003
	(0.007)	(0.007)	(0.008)	(0.009)	(0.009)	(0.010)
Number of Peace Yrs.	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
RQ Ethn. Pol.			-0.004			-0.013
			(0.012)			(0.015)
Excluded Pop. (log)			0.021**			0.024**
			(0.009)			(0.011)
Number Ethn. Grps. in Power			0.002			0.002
			(0.001)			(0.002)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	4330	4330	4330	3583	3583	3583

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (ethnic conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 3: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality': **COUNTRY level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **NON STATE**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.091	0.092	0.073			
	(0.069)	(0.070)	(0.066)			
Gini (rural)				0.150	0.152	0.135
				(0.080)	(0.080)	(0.079)
Drought (% prev. year)	-0.231	-0.234	-0.233	-0.240	-0.242	-0.242
	(0.140)	(0.140)	(0.142)	(0.145)	(0.145)	(0.148)
Precipitation (yearly mean, lagged)	-0.000**	-0.000**	-0.000	-0.000**	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.007**	0.007**	0.008***	0.007**	0.007**	0.008***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Precipitation (deviation, lagged)	0.001	0.004	0.004	0.001	0.009	0.007
	(0.006)	(0.018)	(0.018)	(0.006)	(0.024)	(0.024)
Temperature (deviation, lagged)	-0.006	0.013	0.010	-0.006	0.016	0.012
	(0.005)	(0.019)	(0.018)	(0.005)	(0.018)	(0.017)
Gini (all)*Precipitation (dev.)		-0.006	-0.009			
		(0.027)	(0.027)			
Gini (all)*Temperature (dev.)		-0.030	-0.025			
		(0.029)	(0.028)			
Gini (rural)*Precipitation (dev.)					-0.013	-0.013
					(0.035)	(0.034)
Gini (rural)*Temperature (dev.)					-0.034	-0.028
					(0.027)	(0.026)
Water Coverage (% terr.)	-0.003	-0.003	-0.003	-0.002	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Mountains (% terr.)	0.013	0.014	0.017	-0.001	-0.001	0.006
	(0.069)	(0.069)	(0.065)	(0.067)	(0.067)	(0.063)
Gdp per capita (log, lagged)	0.023**	0.023**	0.018	0.025**	0.025**	0.020
	(0.011)	(0.011)	(0.012)	(0.012)	(0.012)	(0.013)
Agriculture (% Gdp)	0.002	0.002	0.002	0.002	0.002	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Population (log, lagged)	0.059	0.059	0.054	0.058	0.058	0.057
	(0.056)	(0.056)	(0.055)	(0.056)	(0.056)	(0.055)
Education (log)	0.012	0.011	0.009	0.013	0.013	0.011
	(0.009)	(0.008)	(0.008)	(0.009)	(0.009)	(0.008)
Oil (log)	-0.002	-0.002	-0.003	-0.002	-0.002	-0.003
	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)	(0.004)
Trade (log)	0.015	0.015	0.014	0.013	0.014	0.013
	(0.010)	(0.010)	(0.010)	(0.011)	(0.011)	(0.011)
Autocracy	0.017	0.017	0.028	0.025	0.025	0.034
	(0.023)	(0.023)	(0.026)	(0.023)	(0.023)	(0.027)
Number of Peace Yrs.	-0.002**	-0.002**	-0.002	-0.002**	-0.002**	-0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.076			-0.081
			(0.064)			(0.064)
Excluded Pop. (log)			0.085			0.080
			(0.045)			(0.045)
Number Ethn. Grps. in Power			0.008			0.006
			(0.006)			(0.006)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	3270	3270	3270	3218	3218	3218

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (non-state conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 4: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **REGIONAL level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ALL**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.179	0.180**	0.174			
	(0.092)	(0.092)	(0.093)			
Gini (rural)				0.284***	0.287***	0.286***
				(0.110)	(0.111)	(0.108)
Drought (% prev. year)	0.515***	0.514***	0.486***	0.669***	0.670***	0.635***
	(0.189)	(0.187)	(0.185)	(0.225)	(0.223)	(0.222)
Precipitation (yearly mean, lagged)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.005	0.005	0.006	0.007	0.007	0.008**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Precipitation (deviation, lagged)	0.017***	-0.023	-0.022	0.020***	-0.021	-0.021
	(0.007)	(0.023)	(0.023)	(0.008)	(0.025)	(0.025)
Temperature (deviation, lagged)	-0.006	0.038	0.034	-0.009	0.032	0.027
	(0.006)	(0.024)	(0.024)	(0.007)	(0.025)	(0.025)
Gini (all)*Precipitation (dev.)		0.073	0.070			
		(0.040)	(0.040)			
Gini (all)*Temperature (dev.)		-0.079	-0.072			
		(0.044)	(0.043)			
Gini (rural)*Precipitation (dev.)					0.075	0.074
					(0.045)	(0.044)
Gini (rural)*Temperature (dev.)					-0.073	-0.065
					(0.046)	(0.046)
Water Coverage (% terr.)	-0.001	-0.001	-0.000	-0.002	-0.002	-0.001
	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)
Mountains (% terr.)	0.032	0.032	0.036	0.066	0.065	0.078
	(0.067)	(0.067)	(0.065)	(0.086)	(0.085)	(0.081)
Gdp per capita (log, lagged)	0.006	0.006	0.003	0.011	0.011	0.007
	(0.011)	(0.011)	(0.012)	(0.013)	(0.013)	(0.014)
Agriculture (% Gdp)	0.001	0.001	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Population (log, lagged)	0.143***	0.143***	0.122**	0.164**	0.164**	0.163**
	(0.055)	(0.055)	(0.057)	(0.064)	(0.064)	(0.068)
Education (log)	0.011	0.011	0.009	0.012	0.012	0.011
	(0.010)	(0.010)	(0.010)	(0.012)	(0.012)	(0.011)
Oil (log)	0.010	0.009	0.009	0.011	0.011	0.010
	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.006)
Trade (log)	0.011	0.011	0.012	0.008	0.008	0.009
	(0.012)	(0.012)	(0.013)	(0.014)	(0.014)	(0.014)
Autocracy	0.030	0.030	0.026	0.028	0.027	0.023
	(0.033)	(0.033)	(0.037)	(0.037)	(0.037)	(0.043)
Number of Peace Yrs.	-0.007***	-0.007***	-0.007***	-0.007***	-0.007***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.030			-0.079
			(0.075)			(0.097)
Excluded Pop. (log)			0.115			0.119
			(0.061)			(0.070)
Number Ethn. Grps. in Power			0.004			0.002
			(0.007)			(0.008)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	3920	3920	3920	3281	3281	3281

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (all types of conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 5: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **REGIONAL level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ETHNIC**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.039	0.040	0.032			
	(0.023)	(0.022)	(0.023)			
Gini (rural)				0.059**	0.060**	0.050
				(0.027)	(0.026)	(0.027)
Drought (% prev. year)	-0.080	-0.069	-0.069	-0.074	-0.056	-0.062
	(0.060)	(0.059)	(0.059)	(0.071)	(0.067)	(0.068)
Precipitation (yearly mean, lagged)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.000	0.000	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Precipitation (deviation, lagged)	0.001	-0.021***	-0.021**	0.001	-0.026***	-0.025***
	(0.002)	(0.008)	(0.008)	(0.003)	(0.009)	(0.009)
Temperature (deviation, lagged)	-0.001	0.004	0.003	-0.001	0.006	0.005
	(0.002)	(0.011)	(0.012)	(0.003)	(0.014)	(0.014)
Gini (all)*Precipitation (dev.)		0.041***	0.039***			
		(0.014)	(0.015)			
Gini (all)*Temperature (dev.)		-0.009	-0.008			
		(0.019)	(0.019)			
Gini (rural)*Precipitation (dev.)					0.051***	0.049***
					(0.017)	(0.017)
Gini (rural)*Temperature (dev.)					-0.012	-0.011
					(0.024)	(0.024)
Water Coverage (% terr.)	-0.000	-0.000	-0.000	-0.001	-0.001	-0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Mountains (% terr.)	0.003	0.003	0.005	0.009	0.010	0.013
	(0.013)	(0.012)	(0.012)	(0.016)	(0.016)	(0.016)
Gdp per capita (log, lagged)	-0.000	-0.000	-0.002	0.001	0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Agriculture (% Gdp)	0.001***	0.001***	0.001**	0.001**	0.001**	0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Population (log, lagged)	0.034**	0.032**	0.025	0.036**	0.034**	0.029
	(0.015)	(0.014)	(0.015)	(0.016)	(0.016)	(0.017)
Education (log)	0.005**	0.005**	0.004	0.006**	0.006**	0.004
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Oil (log)	0.003***	0.003***	0.003***	0.003***	0.003***	0.003**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Trade (log)	-0.002	-0.001	-0.001	-0.002	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Autocracy	-0.000	-0.001	0.003	-0.003	-0.003	0.001
	(0.008)	(0.009)	(0.010)	(0.010)	(0.011)	(0.012)
Number of Peace Yrs.	-0.001***	-0.001***	-0.001***	-0.001**	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
RQ Ethn. Pol.			-0.004			-0.013
			(0.013)			(0.017)
Excluded Pop. (log)			0.020**			0.022**
			(0.009)			(0.010)
Number Ethn. Grps. in Power			0.002			0.002
			(0.001)			(0.002)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	3920	3920	3920	3281	3281	3281

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (ethnic conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 6: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **REGIONAL level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **NON STATE**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.105	0.105	0.097			
	(0.079)	(0.079)	(0.076)			
Gini (rural)				0.178**	0.180**	0.176**
				(0.090)	(0.091)	(0.090)
Drought (% prev. year)	-0.129	-0.131	-0.135	-0.127	-0.124	-0.134
	(0.137)	(0.134)	(0.137)	(0.135)	(0.132)	(0.137)
Precipitation (yearly mean, lagged)	-0.000**	-0.000**	-0.000	-0.000**	-0.000**	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.007**	0.007**	0.008**	0.007**	0.007**	0.008**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Precipitation (deviation, lagged)	-0.000	-0.009	-0.007	-0.000	-0.019	-0.017
	(0.006)	(0.019)	(0.019)	(0.006)	(0.020)	(0.020)
Temperature (deviation, lagged)	-0.007	0.017	0.015	-0.006	0.017	0.014
	(0.005)	(0.017)	(0.016)	(0.005)	(0.016)	(0.015)
Gini (all)*Precipitation (dev.)		0.015	0.008			
		(0.032)	(0.032)			
Gini (all)*Temperature (dev.)		-0.043	-0.039			
		(0.028)	(0.027)			
Gini (rural)*Precipitation (dev.)					0.034	0.028
					(0.035)	(0.035)
Gini (rural)*Temperature (dev.)					-0.042	-0.037
					(0.026)	(0.025)
Water Coverage (% terr.)	-0.004	-0.004	-0.004	-0.003	-0.003	-0.004
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Mountains (% terr.)	-0.000	-0.000	0.010	-0.024	-0.024	-0.010
	(0.075)	(0.075)	(0.072)	(0.074)	(0.074)	(0.070)
Gdp per capita (log, lagged)	0.022**	0.022**	0.018	0.021	0.021	0.017
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.012)
Agriculture (% Gdp)	0.002	0.002	0.002	0.002	0.002	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Population (log, lagged)	0.057	0.058	0.054	0.063	0.063	0.065
	(0.063)	(0.063)	(0.060)	(0.062)	(0.062)	(0.059)
Education (log)	0.011	0.011	0.008	0.013	0.013	0.011
	(0.009)	(0.009)	(0.009)	(0.010)	(0.009)	(0.009)
Oil (log)	0.000	0.000	-0.001	0.000	0.000	-0.001
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Trade (log)	0.016	0.016	0.015	0.015	0.015	0.014
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Autocracy (d)	0.024	0.025	0.035	0.033	0.033	0.041
	(0.025)	(0.025)	(0.028)	(0.025)	(0.025)	(0.029)
Number of Peace Yrs.	-0.002	-0.002	-0.002	-0.002**	-0.002	-0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.082			-0.089
			(0.071)			(0.071)
Excluded Pop.			0.082			0.078
			(0.049)			(0.050)
Number Ethn. Grps. in Power			0.007			0.006
			(0.006)			(0.006)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	2985	2985	2985	2939	2939	2939

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (non-state conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 7: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **ETHNIC level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ALL**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.258**	0.258**	0.267**			
	(0.116)	(0.116)	(0.115)			
Gini (rural)				0.140	0.140	0.159
				(0.143)	(0.142)	(0.140)
Drought (% prev. year)	0.546**	0.552**	0.492**	0.738***	0.748***	0.683**
	(0.219)	(0.221)	(0.214)	(0.281)	(0.284)	(0.277)
Precipitation (yearly mean, lagged)	-0.001***	-0.001***	-0.001**	-0.001***	-0.001***	-0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.005	0.005	0.007*	0.008*	0.008*	0.011**
	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)
Precipitation (deviation, lagged)	0.022***	0.014	0.013	0.023**	0.017*	0.017*
	(0.008)	(0.008)	(0.008)	(0.009)	(0.010)	(0.010)
Temperature (deviation, lagged)	-0.001	-0.000	0.000	-0.004	-0.006	-0.005
	(0.007)	(0.011)	(0.011)	(0.008)	(0.012)	(0.012)
Gini (all)*Precipitation (dev.)		0.051	0.046			
		(0.041)	(0.040)			
Gini (all)*Temperature (dev.)		-0.002	-0.004			
		(0.043)	(0.042)			
Gini (rural)*Precipitation (dev.)					0.043	0.031
					(0.045)	(0.044)
Gini (rural)*Temperature (dev.)					0.014	0.004
					(0.053)	(0.051)
Water Coverage (% terr.)	0.001	0.001	0.002	-0.000	-0.000	0.001
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
Mountains (% terr.)	0.091	0.090	0.107	0.124	0.124	0.165*
	(0.077)	(0.077)	(0.073)	(0.103)	(0.103)	(0.096)
Gdp per capita (log, lagged)	0.007	0.007	-0.000	0.015	0.015	0.004
	(0.014)	(0.014)	(0.015)	(0.017)	(0.017)	(0.018)
Agriculture (% Gdp)	0.002	0.002	0.001	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Population (log, lagged)	0.052	0.052	0.018	0.057	0.057	0.022
	(0.086)	(0.086)	(0.075)	(0.107)	(0.107)	(0.099)
Education (log)	0.008	0.008	0.006	0.006	0.006	0.004
	(0.012)	(0.012)	(0.011)	(0.014)	(0.014)	(0.013)
Oil (log)	0.012*	0.012*	0.011	0.011	0.011	0.009
	(0.006)	(0.006)	(0.007)	(0.008)	(0.008)	(0.008)
Trade (log)	0.021	0.022	0.021	0.018	0.018	0.017
	(0.014)	(0.014)	(0.014)	(0.016)	(0.016)	(0.016)
Autocracy	0.021	0.022	0.032	0.005	0.006	0.019
	(0.042)	(0.042)	(0.047)	(0.047)	(0.047)	(0.054)
Number of Peace Yrs.	-0.008***	-0.008***	-0.007***	-0.008***	-0.008***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.127			-0.190
			(0.089)			(0.119)
Excluded Pop. (log)			0.194**			0.235**
			(0.075)			(0.095)
Number Ethn. Grps. in Power			0.010			0.012
			(0.008)			(0.009)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	3082	3082	3082	2529	2529	2529

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (all types of conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 8: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **ETHNIC level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **ETHNIC**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.058**	0.058**	0.062**			
	(0.026)	(0.026)	(0.024)			
Gini (rural)				0.089***	0.089***	0.091***
				(0.029)	(0.029)	(0.028)
Drought (% prev. year)	-0.079	-0.076	-0.080	-0.080	-0.070	-0.085
	(0.077)	(0.076)	(0.079)	(0.092)	(0.090)	(0.096)
Precipitation (yearly mean, lagged)	-0.000	-0.000	-0.000	-0.000*	-0.000*	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.000	0.000	0.000	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Precipitation (yearly mean, lagged)	0.003	0.001	0.000	0.004	-0.001	-0.001
	(0.003)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Temperature (yearly mean, lagged)	-0.001	0.001	0.000	-0.000	0.001	0.001
	(0.003)	(0.004)	(0.004)	(0.003)	(0.005)	(0.005)
Gini (all)*Precipitation (dev.)		0.011	0.012			
		(0.017)	(0.016)			
Gini (all)*Temperature (dev.)		-0.009	-0.007			
		(0.016)	(0.016)			
Gini (rural)*Precipitation (dev.)					0.025	0.022
					(0.022)	(0.021)
Gini (rural)*Temperature (dev.)					-0.007	-0.007
					(0.018)	(0.017)
Water Coverage (% terr.)	-0.000	-0.000	0.000	-0.000	-0.000	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Mountains (% terr.)	-0.006	-0.005	0.001	-0.005	-0.005	0.007
	(0.017)	(0.017)	(0.016)	(0.022)	(0.022)	(0.023)
Gdp per capita (log, lagged)	0.001	0.001	-0.001	0.003	0.003	0.000
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Agriculture (% Gdp)	0.001***	0.001***	0.001***	0.001**	0.001**	0.001*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Population (log, lagged)	0.021	0.021	0.007	0.015	0.014	0.003
	(0.023)	(0.023)	(0.022)	(0.028)	(0.028)	(0.028)
Education (log)	0.008**	0.008**	0.006**	0.010**	0.010**	0.008**
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
Oil (log)	0.003**	0.003**	0.003**	0.003**	0.003**	0.003*
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Trade (log)	-0.002	-0.002	-0.002	-0.004	-0.004	-0.004
	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
Autocracy	-0.000	-0.000	0.007	-0.005	-0.004	0.003
	(0.011)	(0.011)	(0.012)	(0.013)	(0.013)	(0.015)
Number of Peace Yrs.	-0.001***	-0.001***	-0.001***	-0.001**	-0.001**	-0.001**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
RQ Ethn. Pol.			-0.017			-0.034
			(0.020)			(0.024)
Excluded Pop. (log)			0.030***			0.035***
			(0.011)			(0.012)
Number Ethn. Grps. in Power			0.004**			0.004
			(0.002)			(0.002)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	3082	3082	3082	2529	2529	2529

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (ethnic conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Table 9: Probability of conflict onset (marginal effects). Gini Index of agricultural (crop) inequality: **ETHNIC level; Models 1-3: all cells, Models 4-6: only rural cells.**

Pr(conflict onset - **NON STATE**)	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gini (all)	0.027	0.027	0.055			
	(0.079)	(0.079)	(0.071)			
Gini (rural)				-0.068	-0.068	-0.035
				(0.091)	(0.090)	(0.086)
Drought (% prev. year)	-0.155	-0.155	-0.167	-0.161	-0.163	-0.184
	(0.165)	(0.165)	(0.149)	(0.175)	(0.174)	(0.160)
Precipitation (yearly mean, lagged)	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Temperature (yearly mean, lagged)	0.008**	0.008**	0.010***	0.007**	0.007**	0.009***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Precipitation (yearly mean, lagged)	0.008	0.009	0.007	0.008	0.011	0.010
	(0.006)	(0.008)	(0.008)	(0.006)	(0.008)	(0.008)
Temperature (yearly mean, lagged)	-0.005	-0.005	-0.004	-0.004	-0.006	-0.004
	(0.005)	(0.008)	(0.008)	(0.006)	(0.008)	(0.007)
Gini (all)*Precipitation (dev.)		-0.004	-0.004			
		(0.031)	(0.031)			
Gini (all)*Temperature (dev.)		0.002	-0.005			
		(0.032)	(0.031)			
Gini (rural)*Precipitation (dev.)					-0.024	-0.026
					(0.034)	(0.034)
Gini (rural)*Temperature (dev.)					0.010	-0.001
					(0.040)	(0.038)
Water Coverage (% terr.)	-0.003	-0.003	-0.002	-0.003	-0.003	-0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Mountains (% terr.)	0.095	0.095	0.113*	0.086	0.086	0.105
	(0.075)	(0.075)	(0.067)	(0.079)	(0.079)	(0.070)
Gdp per capita (log, lagged)	0.021*	0.021*	0.013	0.024*	0.024*	0.015
	(0.012)	(0.012)	(0.013)	(0.012)	(0.012)	(0.014)
Agriculture (% Gdp)	0.002*	0.002*	0.002	0.003*	0.003*	0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Population (log, lagged)	-0.018	-0.018	-0.048	-0.007	-0.007	-0.040
	(0.080)	(0.080)	(0.068)	(0.080)	(0.080)	(0.071)
Education (log)	0.003	0.003	-0.000	0.002	0.002	-0.000
	(0.009)	(0.009)	(0.007)	(0.009)	(0.009)	(0.008)
Oil (log)	0.002	0.002	-0.000	0.003	0.003	0.001
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Trade (log)	0.030***	0.030***	0.027**	0.029***	0.029***	0.026**
	(0.010)	(0.010)	(0.011)	(0.010)	(0.010)	(0.010)
Autocracy	0.017	0.017	0.041	0.022	0.022	0.044
	(0.030)	(0.030)	(0.032)	(0.030)	(0.030)	(0.033)
Number of Peace Yrs.	-0.002	-0.002	-0.001	-0.002*	-0.002*	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
RQ Ethn. Pol.			-0.116			-0.109
			(0.071)			(0.076)
Excluded Pop. (log)			0.131***			0.136***
			(0.050)			(0.051)
Number Ethn. Grps. in Power			0.012**			0.012**
			(0.006)			(0.006)
Geo Macro Area	Yes	Yes	Yes	Yes	Yes	Yes
Time Trend	Yes	Yes	Yes	Yes	Yes	Yes
N. Observations	2456	2456	2456	2366	2366	2366

Notes: *Significance Level:* $*p < 0.1, **p < 0.05, ***p < 0.01$. *The dependent variable is conflict onset (non-state conflicts). The method of estimation is Probit. Robust standard errors are clustered at the country level. Reference categories for dichotomous variables: Democratic politic systems.*

Appendix

Table 10: Summary statistics: 1981 - 2017

	Mean	Std. Dev.	Min.	Max.	N
Dependent Variables					
Onset (all conflicts)	0.224	0.417	0	1	4646
Onset (ethnic conflicts)	0.022	0.146	0	1	4646
Onset (non-state conflicts)	0.077	0.267	0	1	4646
Inequality Indicators					
Gini (country level, all cells)	0.623	0.167	0.014	0.953	4646
Gini (country level, rural cells)	0.634	0.16	0.011	0.952	3862
Gini (regional level, all cells)	0.548	0.149	0.014	0.929	4270
Gini (regional level, rural cells)	0.541	0.148	0.022	0.927	3588
Gini (ethnic level, all cells)	0.167	0.147	0.011	0.67	2938
Gini (ethnic level, rural cells)	0.175	0.149	0.01	0.653	2420
Climatic Variables					
Precipitation (yearly mean)	86.56	72.070	0.003	583.791	4605
Temperature (yearly mean)	18.691	8.063	-7.426	29.541	4295
Drought (% prev. year)	0.056	0.042	0	0.354	4646
Precipitation (deviation)	0.008	0.99	-3.69	3.708	4605
Temperature (deviation)	0.015	0.985	-5.317	4.222	4646
Country Characteristics					
Water Coverage (% territory)	6.646	7.707	0	56.795	4646
Mountains (share territory)	0.314	0.27	0	0.961	4646
Gdp per capita (log)	15.581	2.132	10.199	20.908	4646
Population (log)	0.782	0.351	0	1	4646
Education (log)	15.438	1.608	9.324	20.768	4646
Oil (log)	20.921	3.026	9.76	27.257	4646
Trade (log)	24.786	1.857	16.826	29.807	4646
Autocracy	0.465	0.499	0	1	4646
Deocracy	0.535	0.499	0	1	4646
Agriculture (% Gdp)	16.323	14.191	0.094	81.714	4646
RQ Ethn. Pol.	0.514	0.318	0	0.995	4646
Excluded Population (log)	0.166	0.222	0	0.92	4646
Number Ethnic Groups in Power	1.803	2.062	0	15	4646
Number Peace Years (ref. year: 1946)	21.486	20.189	0	71	4646
Other Controls					
North America	0.023	0.15	0	1	4646
Central America	0.044	0.204	0	1	4646
Latin America	0.117	0.322	0	1	4646
Western Europe	0.139	0.346	0	1	4646
Eastern Europe	0.089	0.285	0	1	4646
North Africa and Middle East	0.121	0.327	0	1	4646
Sub-Saharan Africa	0.269	0.444	0	1	4646
Asia	0.168	0.374	0	1	4646
Oceania	0.029	0.168	0	1	4646

Table 11: Definition of the main variables included in the analysis and corresponding data-sources.

VARIABLE	DEFINITION	DATA SOURCE
Onset of civil conflict	Binary variable coded as 1 for each new intra-state conflict out breaking in a cell-year and 0 otherwise. Intra-state conflict involves the use of the armed force by either the government or a formally organized group, and results in at least 25 battle-related deaths in a year.	UCDP-GED (Sundberg and Melander, 2013).
Onset of ethnic conflict	Binary variable coded as 1 for each new intra-state ethnic conflict out breaking in a cell-year and 0 otherwise.	GeoEPR 2019 (Voigt et al., 2015).
Population	Population size for each country.	GeoEPR 2019 (Vogt et al., 2015).
Excluded groups	Number of excluded groups (discriminated or powerless) as defined in the GeoEPR/EPR data on the status and location of politically relevant ethnic groups settled in the grid cell for the given year.	GeoEPR (Vogt et al., 2015).
GDP	Total GDP (purchasing power parity of each 0.5x0.5 grid cell, derived from GDP per capita (PPP) which is multiplied by gridded population data.	Gridded global datasets for Gross Domestic Product and Human Development Index (Kummu et al., 2018).
GDP per capita	Gridded GDP per capita, derived from a combination of sub-national and national datasets.	Gridded global datasets for Gross Domestic Product and Human Development Index (Kummu et al., 2018).
Oil rents	Difference between the value of crude oil production at world prices and total costs of production.	World Bank Development Indicator (World Bank, 2012).
Trade	Trade volume.	World Bank Development Indicator (World Bank, 2012).
Education	Number of people aged over 15 who can read expressed as share of the total population.	World Bank Development Indicator (World Bank, 2012).

VARIABLE	DEFINITION	DATA SOURCE
Autocracy	Variable identifying autocracies. The classification is performed according to V-Dem Indices of political systems and corresponds to regime types 0 and 1 (closed autocracy).	Coppedge et al., 2019.
History of conflict	Binary variable coded as 1 if the grid cell experienced a conflict in the previous year, and 0 otherwise.	Our computation from UCDP-GED (Sundberg and Melander, 2013).
Number of peace years	Number of years since the last conflict experienced by the country.	UCDP-GED (Sundberg and Melander, 2013).
Mountainous terrain	Proportion of mountainous terrain within the cell based on elevation, slope and local elevation range, taken from high-resolution mountain raster data.	UNEP's Mountain Watch Report (Blyth et al., 2003).
Share of the Excl. Pop.	Indicator of the degree of exclusion along ethnic lines, computed as the natural logarithm of the share of the population excluded from central government.	Voigt et al., 2015.
Num. Power Shar Part.	Number of power sharing groups represented by ethnic elites at the central government. The variable is termed as the degree of center segmentation.	Voigt et al., 2015.
Ethnic Polarization	Reynal-Querol discrete polarization index, measuring the degree of ethnic polarization within countries.	Montalvo and Reynal-Querol, 2002, 2005.
Crop	Annual production of four main crop types within the 0.5x0.5Â° grid cell (maize, soybean, wheat, rice) given in tons per hectare.	Iizumi et al., 2018.
Water (gc)	Coverage of water areas in each cell.	Globcover v2.3. (Bontemps et al., 2009)
Agr. value added	Net output of the agricultural sector as a share of the national GDP. Agriculture includes forestry, hunting, and fishing, as well as cultivation of crops and livestock production.	World Bank Development Indicators (World Bank, 2012)

VARIABLE	DEFINITION	DATA SOURCE
Drought (crop)	Proportion of months in the growing season that are part of the longest streak of consecutive months in that growing season with SPEI values below -1.5. The growing season is the growing season for the cell's main crop, defined in the MIRCA2000 dataset v.1.1.	SPEI Index (Begueria et al., 2014).
Precipitation anomaly	Precipitation anomaly, computed as the difference of the annual precipitation mean from the long-term climate mean.	University of East Anglia Climatic Research Unit (CRU), 2017.
Temperature anomaly	Temperature anomaly, computed as the difference of the annual precipitation mean from the long-term climate mean.	University of East Anglia Climatic Research Unit (CRU), 2017.